TeduhLink

A connected rider-stop network

Team discussion draft | Revised concept | 24 October 2026

> **The question we help a rider answer:** Which participating stop can meet my needs right now?

The idea in everyday language

A rider needs somewhere to stop between deliveries or during bad weather. TeduhLink would show participating rest points, explain how to access them, and combine their listed amenities with current information from connected devices.

The rider could check whether a monitored space is occupied, which information is current, and which suitable option is nearby. The aim is to reduce time spent looking for a stop and avoid journeys to unsuitable facilities.

Those are intended benefits to test. We have not yet established how often Cyberjaya riders experience this information problem.

A quick example

A rider checks the service while stationary and asks for shelter and phone charging. Stop A is closer but its monitored space is occupied. Stop B has suitable listed amenities and a current available-space reading. TeduhLink suggests B and explains the reason.

If B stops reporting, its live status becomes unknown. The service should show that uncertainty rather than continue presenting the old reading as current.

What is different in this version?

The main service is finding a suitable rider stop using reliable information. The earlier junction-warning and Safe Approach modules are outside this version's core scope.

Existing TEDUH facilities provide the physical shelter and amenities. Our proposed addition is information and coordination before arrival.

What this document is for

Read the concept, understand the proposed prototype and discuss how to improve it. The final page contains questions to answer together before committing to the build.

> **Decision still open:** Can we establish a real rider information need, and is our proposed 5G contribution strong enough for this competition?

1. The problem and the local evidence

What we can support now

An official MPHS report records a 4 May 2026 visit to MPSepang's TEDUH project in Cyberjaya. It establishes a local initiative we can investigate; it does not confirm today's operating hours, capacity or digital systems. [1]

Media Selangor reported in September 2026 that TEDUH facilities were available across Selangor's 12 municipalities. It describes rest between orders and shelter from heat or rain. This supports the wider service context. [2]

The additional problem we must validate

A facility existing does not automatically mean riders have difficulty finding or using it. We need evidence for at least one specific gap:

- Riders cannot easily identify a suitable participating stop.
- Access details, opening arrangements or amenities are unclear before arrival.
- Riders arrive to find a needed space or amenity unavailable.
- Operators cannot keep useful facility information current.

These are research questions, not findings. Do not state that Cyberjaya TEDUH is crowded, underused or poorly maintained without evidence.

Who would benefit?

Primary users: Riders looking for an eligible place to rest or wait during their working day.

Potential operator: A council, facility manager or participating host responsible for listing and maintaining accurate information.

Potential later users: Additional rider groups or the wider public only where facility access rules allow them.

Our working problem statement

"We will investigate whether riders in Cyberjaya have difficulty identifying an accessible rest point that meets their needs at the time they need it. TeduhLink proposes a shared information service combining verified facility details with current readings from monitored spaces."

A credible first validation

Seek a few concrete accounts: the last time a rider needed a stop, how they found it, what made it suitable or unsuitable, and what information would have changed their decision. Record contrary answers too.

> A few specific experiences can guide a pilot, but they do not establish citywide demand. If riders say existing information is sufficient, revise the proposed service.

2. What the rider sees and does

One short journey through the service

1. **Choose a need while stationary.** Examples: a covered rest point or a listed phone-charging amenity.
2. **See participating options.** Each has access information, amenities, the monitored-space status and the latest update time.
3. **Understand the suggestion.** The system explains which stated requirements are met and what remains unknown.
4. **Choose a destination.** The rider follows verified access instructions or an existing mapping service. TeduhLink does not certify a safe route.
5. **Check the current information.** Availability is an observation, not a reservation; another rider may arrive first.

STOP A — PHYSICAL MODEL

Listed: shelter + charging amenity

Model bay: occupied

Current test reading

STOP B — SIMULATED STOP

Listed: shelter + charging amenity

Simulated space: available

Current simulated update

Example: suggest B because A has no available monitored space.

Illustrative test data; these are not confirmed participating facilities.

How the recommendation works

Start with clear rules. Filter options using access eligibility and the rider's required amenities. Use relevant current capacity information where available, then compare suitable options by a disclosed distance estimate or preference.

Do not quietly treat unknown information as a confirmed match. If no option has enough information to meet the request, say so and show which facts are missing.

Use precise status labels

- **Listed:** An amenity recorded in the facility catalogue, with its verification date.
- **Measured:** A device observation about a named bay or space, with its timestamp.
- **Unknown:** Missing, expired or unverified information.
- **Simulated:** A test value generated for the demonstration.

A reading for one monitored bay must not be presented as the whole facility's capacity. Charging being listed does not prove that a particular socket is currently working.

> **Core value to test:** Can the live information help someone choose a suitable stop more effectively than a static map or facility list?

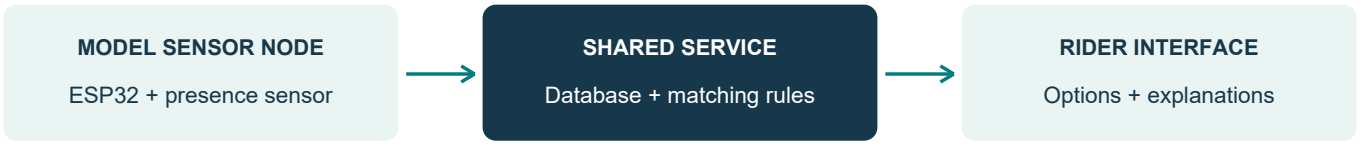
3. IoT, AI and 5G: each has a purpose

IoT provides current observations

For the minimum prototype, use an ESP32 and a suitable presence sensor to monitor one model bay. It sends a bay ID, occupancy state, timestamp and device status.

Use a small object to simulate occupancy. This demonstrates the data flow; it does not validate detection of real riders, seats or motorcycles outdoors.

Keep charging as clearly labelled catalogue or simulated information initially. USB voltage or current alone does not establish that a charging point is functional and available. Add a real charger-status feature only with a defined, tested measurement.



The phone / router carries node updates over cellular to the external service.
One physical model node + one clearly labelled simulated second stop.

AI / Computer Science builds the useful service

The member owns the recommendation rules, database, interface, explanations and tests. These are substantial software tasks even when a trained AI model is unnecessary.

An optional AI feature could estimate occupancy from an appropriate camera view. It should solve a need the simple sensor cannot meet, and its accuracy must be evaluated separately.

GenAI could explain recorded information in Bahasa Melayu or English with references to the underlying facts. It must acknowledge unknown values. A learned demand forecast is not justified by five days of sparse data.

5G connects the participating nodes

The phone or router provides cellular connectivity; the ESP32 connects to it using Wi-Fi. Messages must cross cellular to an external service if we claim a 5G-connected demonstration.

The unresolved issue: These small messages can also travel over 4G or other networks. A successful upload proves connectivity, not that 5G is necessary.

Measure delivery, stale information and recovery; compare modes fairly if possible. Do not add unnecessary video feeds just to create demand for 5G. The team must decide whether its evidence supports a material benefit.

> Track 2 can include IoT. Choose the track by the rider and mobility problem; the website's example technologies are not stated as exclusive to Track 1. [3]

4. What each member would build

Networking member / project lead

Own the message route, shared data format, authentication, timestamps, expiry and reconnection. Verify the actual cellular path and collect delivery results. Coordinate evidence, scope and the final Innovation Canvas.

Deliverable: The device updates reach the shared service; expired information becomes unknown; the network demonstration has measured results.

AI / Computer Science member

Own the database, rider interface, recommendation rules, explanations and evaluation. Keep listed facts, live observations and simulated readings visibly distinct. Add a trained AI feature only after the core works and a useful task is identified.

Deliverable: A rider request produces an understandable suggestion from the available evidence, or clearly explains why no option can be confirmed.

Mechatronics member

Own the model bay, sensor placement, ESP32 firmware, power and calibration. Test occupied, empty and unreliable readings, and report exactly what the sensor measures.

Deliverable: Moving an object into and out of the model bay creates repeatable updates with the agreed status and timestamps.

The interface to agree before coding

Use the same site ID, bay ID, status names and time format. Agree the update frequency, freshness timeout and how an unknown reading is represented. Choose a simple database and software stack the team already knows.

Minimum materials and demonstration scope

One laptop; one ESP32; one suitable presence sensor; breadboard and wires; regulated USB power; a 5G phone / router and data plan; cardboard for one model stop; and an authenticated external service.

Build one physical model node. Represent a second stop openly as simulated. Neither model is an installed TEDUH device or a confirmed partner location. A Raspberry Pi, camera and 3D printer are optional.

> Shared ownership: all three members review the rider evidence, define the benefit test and agree what the team can claim.

5. A small prototype we can finish

Five-day plan

1. **Day 1 - validate and agree.** Review the problem evidence, seek concrete rider accounts, define one request and freeze the data format. Decide how to handle the weak 5G argument.
2. **Day 2 - connect one physical node.** Make the sensor update a database-backed screen through the intended connection. Mark all simulated inputs.
3. **Day 3 - add a second simulated stop.** Implement transparent matching, explanations, update times and unknown status.
4. **Day 4 - test the benefit and failures.** Compare the static listing and live information; test sensor loss, old data and reconnection.
5. **Day 5 - prepare the submission.** Record a short demonstration, complete the canvas and check every claim against its evidence.

The judge demonstration

Show a stationary rider choosing shelter and a listed charging amenity. Start with two clearly labelled test stops. Occupy the physical model bay and show the updated information changing the suggestion.

Disconnect that node. Show unknown status instead of an old availability reading. Reconnect and let fresh information restore the display.

Explain what the suggestion used, which values were simulated and which measurements passed through cellular. This visible response is the proposed demonstration, not proof of real-world demand.

What to measure

- Accuracy on repeated model occupied / empty trials, including unclear readings.
- Time from a reading being generated to a visible update; state the measurement method.
- Correct treatment of expired, missing and restored data.
- Whether task participants choose a suitable stop more effectively with live information than with the same static listing.

For a small task test, use the same defined scenarios, alternate presentation order and report actual outcomes with sample sizes. Do not claim reduced accidents or citywide time savings from a model test.

How it could expand

A later pilot could add authorised TEDUH sites and other eligible hosts. Each needs verified access details, an installation plan, power, maintenance and responsibility for the catalogue. A network requires participating facilities; map pins alone do not create one.

> The public challenge site says a prototype is optional. Protect time for the researched problem, clear concept, 5G explanation and feasibility plan. [3]

6. Questions for our team discussion

Discuss these before we commit

1. **Which single rider difficulty are we solving?** What actual account or observation supports it?
2. **What must the rider know before arrival?** Which information would change a decision?
3. **What can our hardware establish reliably?** Does it report one bay, an amenity or the whole facility?
4. **What is better than a static listing?** Which comparison could demonstrate that benefit?
5. **Does AI add useful capability?** Which task needs a trained model, if any?
6. **Why is 5G materially useful?** What can we measure, and what remains an assumption?
7. **Who keeps the service accurate?** Who approves listings, maintains sensors and responds to faults?
8. **What do we remove to finish in five days?** What is the smallest complete demonstration?

Write the decision together

Main rider need: _____

Evidence still needed / person responsible: _____

Minimum feature to demonstrate: _____

Most important improvement suggested: _____

Next action, owner and completion time: _____

Decision: continue / revise / investigate another concept

Sources and boundaries

[1] MPHS, "Lawatan Kerja MPHS ke Projek Teduh MP Sepang," 4 May 2026. Supports the local TEDUH initiative, not current availability or an information gap.
https://www.mphs.gov.my/ms/mphs/media/arkib/arkib-berita/lawatan-kerja-mphs-ke-projek-teduh-mp-sepang

[2] Media Selangor, "TEDUH p-hailing rest stops now available in all 12 Selangor municipalities," 3 September 2026. Statewide programme context. https://mediaselangor.com/en/2026/09/393793

[3] Official Cyberjaya 5G challenge website. Track examples, concept focus and optional prototype.
https://cyberjaya5g.kolaxus.net/

Sources reviewed 24 October 2026. This is a student discussion concept. Operator participation, rider demand, measured benefits and a material 5G advantage remain to be established.